Offroad Segmentation

Complete Setup Guide — All Hardware Tiers

BigRock Hackathon

⚠️ **START TRAINING THE MOMENT THE HACKATHON BEGINS** — training takes 1–3+ hours. Set it running first, then read this guide while it trains.

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0 Before You Start — Required Accounts & Data

⚠️ Complete **BOTH** steps below before anything else. You need the Falcon account to access the dataset.

Step A — Create your Falcon account

Visit the link below and sign up with your email:

falcon.duality.ai — Sign Up

Field	What to enter
Email	Your college / personal email
Password	Choose a strong password
Name	Your full name
Organisation	Your college / team name

Step B — Download the dataset from Falcon

After signing in, go to the hackathon dataset page:

falcon.duality.ai — Hackathon Dataset

Download **all dataset ZIPs** from this page: Training Dataset, Test Images, and Scripts. Save them all to one folder, e.g. `Downloads/TRACK2/`

1 Minimum System Requirements

Component	Minimum

OS	Windows 10 / Ubuntu 20.04 / macOS 12
RAM	8 GB
Storage	10 GB free
GPU	None (CPU fallback works)
Python	3.10
Internet	Required (model download ~84 MB)

2 All Downloads with Links

Download every item below before starting. All links are direct and free.

#	Item	Link	Notes
1	Training Dataset	Falcon Portal	ZIP from Falcon
2	Test Images	Same page	ZIP from Falcon
3	Scripts ZIP	Same page	train/test .py files
4	Miniconda (Windows)	Download .exe	~70 MB installer
5	Miniconda (macOS Intel)	Download .sh	bash installer
6	Miniconda (Apple Silicon)	Download .sh	M1/M2/M3 Macs
7	Miniconda (Linux)	Download .sh	bash installer
8	VS Code	Download	Recommended editor
9	Google Colab	Open Colab	Free T4 GPU, no install

3 Step-by-Step Setup

Step 1 — Install Miniconda

Windows

Double-click the `.exe` → Next → I Agree → Just Me → on Advanced Options check both boxes:

- ☑ Add Miniconda3 to my PATH
- ☑ Register Miniconda3 as my default Python

Click Install → wait ~2 min → Finish

macOS / Linux

```
bash ~/Downloads/Miniconda3-latest-*.sh
# Follow prompts, type yes when asked
source ~/.bashrc    # or ~/.zshrc on macOS
```

Verify

```
conda --version    # should show conda 24.x.x
```

Step 2 — Create EDU Environment

ℹ️ Cloud users (Colab/Kaggle): Skip this step — use `!pip install` directly in cells.

```
conda create -n EDU python=3.10 -y
conda activate EDU
# Prompt should now show (EDU)
```

Step 3 — Install PyTorch

RTX 40-series (CUDA 12.1)

```
pip install torch torchvision torchaudio --index-url https://download.pyt
```

GTX / RTX 10xx–30xx (CUDA 11.8)

```
pip install torch torchvision torchaudio --index-url https://download.pyt
```

CPU only / iGPU / macOS

```
pip install torch torchvision torchaudio
```

All tiers — supporting libraries

```
pip install opencv-python pillow matplotlib tqdm numpy
```

Step 4 — Verify GPU Detection

```
python -c "import torch; print(torch.cuda.is_available()); \
print(torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'CP
```

GPU users — expected: `True` then your GPU name

CPU users — expected: `False` then `'CPU mode - OK'` — this is normal

Step 5 — Extract Datasets & Scripts

Extract all 3 ZIPs into one folder with this structure:

```
TRACK2/
├── Offroad_Segmentation_Training_Dataset/
│   ├── train/
│   │   ├── Color_Images/
│   │   └── Segmentation/
│   └── val/
│       ├── Color_Images/
│       └── Segmentation/
├── Offroad_Segmentation_testImages/
│   └── Color_Images/
└── Offroad_Segmentation_Scripts/
    ├── train_segmentation.py
    ├── test_segmentation.py
    └── visualize.py
```

Colab: Upload ZIPs to Google Drive, then unzip with:

```
!unzip /content/drive/MyDrive/TRACK2/Offroad_*.zip -d /content/drive/MyDrive/TRACK2/
```

⚡ Step 6 — Replace Script Files & Start Training NOW

⚠️ Do this **immediately** — training takes 1–3+ hours. Kick it off first, then continue with steps 7–8.

Open `train_segmentation.py` in VS Code → select all → paste corrected code (Flowers class, 11 classes, proper value_map). Repeat for `test_segmentation.py`. Save both.

→ No VS Code? Right-click `.py` → Open with Notepad (Win) or TextEdit (macOS) → Select All → paste → save.

```
cd TRACK2/Offroad_Segmentation_Scripts
python train_segmentation.py
```

✅ **Training is running!** DINOv2 backbone downloads ~84 MB on first run. Leave this terminal open and continue reading.

Step 7 — Baseline Training — 10 epochs [optional]

ℹ️ Run only for a quick sanity-check score before investing hours in the full model. Baseline IoU ~0.20.

```
python train_segmentation.py    # default 10 epochs
```

Expected: DINOv2 downloads ~84 MB (one-time).

- ~56 min on RTX 4060
- ~2 hrs on older GPU

- ~6 hrs on CPU

⚠️ **CPU/iGPU users:** set `n_epochs = 5` in `train_segmentation.py` to stay under 5 hours.

Step 8 — Run Test Script [optional]

ℹ️ Run after Step 7 only. Checks predictions on 1002 test images and reports baseline score.

```
python test_segmentation.py
```

Expected: processes 1002 images in ~1–2 min · saves predictions to `predictions/` · Mean IoU ~0.20

Step 10 — Package Submission

Create `TeamName_Submission.zip` with this structure:

```
TeamName_Submission/
├── train_segmentation.py
├── test_segmentation.py
├── segmentation_head.pth
├── train_stats/
├── predictions/
├── README.md
└── Report.pdf
```

4 Recommended 8-Hour Timeline

0 – 1h	Falcon signup · dataset download · Miniconda install · GPU verify · **START TRAINING** (Step 6)	Training running
1 – 4h	Configure 30-epoch improved model (Step 9), let it run	segmentation_head.pth v2
4 – 6h	Write report while training finishes · run test script	Report draft + predictions/
6 – 8h	Final packaging · README · optional 50-epoch run	TeamName_Submission.zip

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Good luck! 🚀 — Generated for BigRock Hackathon participants